

3342 Review: Chapters 4 - 5.3

Terms and Representative Problems

Chapter 4

random variable

probability distribution 2-5

$$f(x) \geq 0$$
$$\sum_{\text{all } x} f(x) = 1$$

discrete random variable

continuous random variable

probability histogram

cumulative distribution function

binomial distribution 13-20

two outcomes per trial

p(success) same for all trials

fixed number, n, of trials

trials are independent

binomial distribution function

$b(x;n,p)$ 7

cumulative binomial distribution function

$B(x;n,p)$

symmetric

positively skewed

negatively skewed

hypergeometric distribution 22-27

sampling without replacement

hypergeometric distribution function

$h(x;n,a,N)$

mean 30, 32

binomial 38-39

hypergeometric

variance and standard deviation 30, 32

binomial

hypergeometric

kth moment about the origin

alternate formula for variance 31, 33

Chebyshev's Theorem 44-45

law of large numbers

Poisson distribution 54-57

mean and variance

approximation to binomial 52-53

Poisson process 63-65

geometric distribution 60, 62

mean and variance

multinomial distribution 70, 72

mean and variance

Chapter 5

probability density function 2, 4, 6, 9-10

$$f(x) \geq 0$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

distribution function 5

kth moment about the origin

mean, variance and standard deviation
13-14

normal distribution 24, 27, 29, 31, 33

mean and variance

standard normal distribution 19-21

Table 3

standardized random variable

normal approximation to binomial 35-39

continuity correction